

CLS Linac Modulator Upgrade Requirements for Toshiba Klystrons

1.8.32.5 Rev 1

Date: 2015-Feb-05

Copyright 2015, Canadian Light Source Inc. This document is the property of Canadian Light Source Inc. (CLSI). No exploitation or transfer of any information contained herein is permitted in the absence of an agreement with CLSI, and neither the document nor any such information may be released without the written consent of CLSI.

Canadian Light Source Inc.
44 Innovation Boulevard
Saskatoon, Saskatchewan
S7N 2V3 Canada

Signature

Date

Original on File – Signed by:

Author	_____ Accelerator Physicist	_____
Reviewer #1	_____ Electrical Engineer	_____
Reviewer #2	_____ Assistant Director of Accelerators	_____
Approver	_____ Director of Accelerator	_____

TABLE OF CONTENTS

1.0	Purpose.....	1
2.0	Scope	1
3.0	General Background	1
4.0	Definitions	2
5.0	Klystron and Modualtor Specifications	2
5.1	ITT klystron 8568 specification	2
5.2	K2 modualtor specification	3
5.3	Toshiba E37300 klystron specification	4
5.4	K2 modulator upgrade specification	6
6.0	Other Modulator K2 Upgrade Requirements	7
6.1	Remove capacitive plate in transformer tank.....	8
6.2	Replace klystron socket to accommodate E37300	8
6.3	Replace the cables between switch units and transformer tank.....	8
6.4	Replace pulse tuning circuits in the oil tank	8
6.5	Replace klystron tank lid	9
6.6	Replace the klystron focus solenoids	9
6.7	Install the solenoid power supplies.....	9
6.8	Replace the CTV boards if necessary	9
6.9	Control system upgrade	10
6.10	Covers	10
6.11	Water system modification	10
6.12	Ion pump high voltage power supply	10
6.13	Scandinova shall be responsible for all necessary upgrades.....	10
7.0	Shipment.....	11
8.0	Applicable Codes, Standards and Procedures.....	11
9.0	Quality Assurance	12
10.0	Design Review, Inspection Testing and Commissioning	12
11.0	Reliability and Maintainability	13
12.0	Other Requirments and Constraints.....	15
13.0	Drawings and Documentation	16
14.0	References.....	16
	Revision History	17

1.0 PURPOSE

This document specifies the required modification to the ScandiNova modulators used at CLS linac to fit the new Toshiba klystron E37300.

2.0 SCOPE

ScandiNova will furnish all equipment, materials, tools, facilities, and labor to perform all the work necessary to design, manufacture, install and test the modulators as per the specifications. This work includes, but is not limited to:

- Materials
- Equipment
- Detailed drawings (if different from the old K2)
- Operation and maintenance manuals (if different from the old K2)
- Fabrication
- Delivery to site
- Assembly
- Testing
- Test report
- Certification and marking acceptable for Canada
- Quality Assurance /Quality Control documentation
- Warranty
- Recommended spare components

3.0 GENERAL BACKGROUND

The Canadian Light Source (CLS) is a national facility operating on the University of Saskatchewan campus in Saskatoon, Saskatchewan. This facility is a 3rd generation synchrotron light source, which produces a high intensity source of infrared, visible, ultraviolet and x-ray radiation.

The facility includes a 250 MeV linear accelerator which uses six ScandiNova K2 modulators to power the klystrons which amplify the RF. A seventh modulator is used for our klystron conditioning/test facility. Presently the existing K2 modulators are designed for ITT 8568 klystrons. However, the company no longer manufactures 8568 klystron. Replacement klystrons have been selected. Toshiba klystron E37300 will be installed over the next three to four years. To fit the E37300 klystron, the existing K2 modulators will need to be modified. CLSI has contracted directly with Toshiba for the purchase of the new klystrons.

As discussed with ScandiNova, the modulator can keep most of the main parts. Only some parts need to be replaced with newly designed or new upgraded parts to accommodate the new klystrons.

4.0 DEFINITIONS

ANSI	American National Standards Institute
CSA	Canadian Standards Association
DCS	Distributed Control System
EPICS	Experimental Physics and Industrial Control System
IEEE	Institute of Electrical and Electronics Engineers
MOV	Metal Oxide Varistor
NEMA	National Electrical Manufacturers Association
PLC	Programmable Logic Controller
RF	Radio Frequency
RMS	Root Mean Square

Shall: Indicates a requirement that is to be strictly followed in order to conform to the specification.

Should: Indicates a recommended or preferred requirement. Deviation must be justified.

May: Indicates a possible method or course of action.

5.0 KLYSTRON AND MODULATOR SPECIFICATIONS

ScandiNova K2 modulators used for the CLS linac presently supply pulse high voltage for ITT 8568 klystrons. Six modulators were installed in March, 2011 and another modulator used for test purpose was installed in the test modulator room in 2010. All of them have been working for three years.

ITT 8568 klystrons are obsolete. Toshiba E37300 klystrons will replace 8568 to supply power to the CLS linac.

K2 modulators cannot work with the new klystron directly. They have to be modified to fit the klystron E37300. For this purpose, all klystrons and modulators specifications have been listed below for reference.

5.1 ITT KLYSTRON 8568 SPECIFICATION

ITT klystron 8568 has been in operation at CLS for fifty years. The klystron specification is listed below only for reference to understand what should be modified in the ScandiNova K2 modulators used at CLS in comparison to the new Toshiba klystron E37300.

Table 1. ITT 8568 Klystron Specification

ITT 8568 Klystron Specification		
Parameters	Range	Unit
Frequency	2856	MHz
Peak RF Output Power	21	MW
Peak RF Drive Power (VSWR<1.5:1)	105	W
Gain	53	dB
Average RF Output Power	≤ 18.8	kW
Peak Forward Beam Voltage	250	kV
Peak Cathode Current	250	A
(Peak Inverse Beam Voltage)	50	kV
Pulsed Peak Power	62.5	MW
Efficiency	33.6	%
Repetition Rate	1-280	Hz
Pulse Width	3.2	μ s
VSWR	1.5:1	
Klystron Filament Voltage	15	V
Klystron Filament Current – *Note 1	14	A
Klystron warm-up time	20	Minutes

Note 1: When the heater power is applied to a cold tube, the heater voltage shall be adjusted from zero to the prescribed value so that the heater current should be kept at 14 A (Not to exceed 14 A! otherwise, the klystron may be damaged). This heater voltage value shall be maintained for at least 20 minutes prior to the application of beam voltage. The liquid coolant flow must be operating whenever the heater power is applied.

5.2 K2 MODULATOR SPECIFICATION

The ScandiNova modulator K2 installed at CLS was originally designed for the ITT klystron 8568. The original modulator specifications are listed as follows:

Table 2. Modulator K2 Specification for ITT 8568

Original ScandiNova K2 Modulator Specification (for klystron 8568)		
Parameters	Range	Unit
Peak Power	62.5	MW
Pulse Voltage	250	kV
Pulse Current	250	A
Average Power	≤ 0.75	kW
Repetition Rate	1-3	Hz
Pulse Width (3 μ s at the flat top)	4	μ s
Voltage Rise Time	1	μ s
Voltage Fall Time	1	μ s
Voltage Top Flatness	± 0.25	%
Voltage Repeatability (pulse to pulse)	± 0.05	%
Pulse Jitter	$\leq \pm 5$	ns
Modulator Efficiency	> 80	%
Klystron Filament Voltage	15	V
Klystron Filament Current	14	A

5.3 TOSHIBA E37300 KLYSTRON SPECIFICATION

The klystron E37300 was proposed by Toshiba to replace ITT klystron 8568 that has been used in CLS's linac for over fifty years.

CLS needs at least 20 MW RF power from a single klystron to drive an acceleration section in normal operation mode. But, due to RF and vacuum conditioning purposes, we need **23-25 MW** RF power in RF conditioning mode.

The klystron E37300 has perveance of $2.2 \mu\text{A/V}^{1.5}$. From the calculated results in table 3 and figure 1, the klystron can supply 22.6 MW RF power with 242 kV high voltage and 262 A beam current that would be the lowest RF conditioning requirements. Klystron specifications are given in table 4.

Table 3. Klystron E37300 RF Power vs. High Voltage

Toshiba Klystron E37300		
High Voltage	Beam Current	Klystron RF Power
(kV)	(A)	(MW)
234	249.0	20.0
238	255.4	21.3
242	261.9	22.6
246	268.4	23.9
250	275.0	25.1

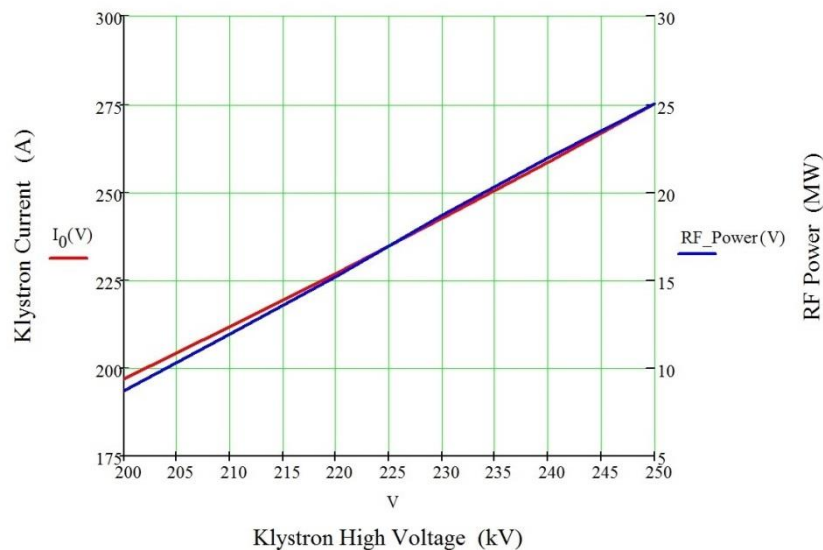


Figure 1. The Klystron E37300 RF Power

Table 4. Klystron E37300 Specification

Toshiba Klystron E37300 Specification		
Parameters	Range	Unit
Frequency	2856	MHz
Peak RF Output Power	25	MW
Peak RF Drive Power	400	W
Gain	48	dB
Average RF Output Power	≤ 1	kW
VSWR (normal load VSWR=1.2:1)	<1.4:1	
Peak Voltage	250	kV
(Peak Inverse Voltage)	100	kV
Peak Current	275	A
Peak Power	68.8	MW
Efficiency	36.4	%
Repetition Rate	1-10	Hz
Pulse Width (RF flat top 4 μ s)	7	μ s
Klystron Filament Voltage	20	V
Klystron Filament Current – *Note 2	20	A
Klystron warm-up time	60	Minutes

Note 2: When the heater power is applied to a cold tube, the heater voltage shall be adjusted from zero to the prescribed value so that the heater current should be kept at

20 A (Not to exceed 40 A! otherwise, the klystron may be damaged). This heater voltage value shall be maintained for at least 60 minutes prior to the application of beam voltage. The liquid coolant flow must be operating whenever the heater power is applied.

5.4 K2 MODULATOR UPGRADE SPECIFICATION

Based on the Toshiba klystron E37300 and the CLS linac operation requirements, the ScandiNova modulator K2 shall satisfy the requirements of the klystron E37300. Details are shown in table 5.

The biggest change will be the klystron current increase from 250 A to 275 A that will need the total power to increase also. The CLS K2 modulator was designed for the ITT 8568 klystron with the repetition rate up to 3 Hz originally. The repetition can be reduced to 2 Hz if necessary when it is upgraded. The average power decrease may help the switch units and the DC power supply to work in the same capacitance.

The important thing is that the high voltage top flatness of the pulse has not met the requirements in the specification since the modulator commissioning three years ago. The reason was the tuning circuits were designed and adjusted only for a klystron that was sent to Uppsala. To other klystrons with slightly different parameters, the modulators with same or similar tuning circuit did not meet the requirements of the flatness. We hope ScandiNova can solve this problem to let the Voltage Top Flatness (in 3 μ s) meet the requirement $\pm 0.25\%$ for all modulators.

Another noticeable point is that the peak inverse beam voltage of the Toshiba E37300 klystron is 100 kV, which is higher than the ITT klystron 8568's 50 kV peak inverse beam voltage.

The klystron filament power increases from 210 Watts (15V @ 14A) up to 400 Watts (20V @ 20A). The filament power supplies in use may satisfy the requirements of the increase.

The other details of the upgrade parts and devices in the modulator K2 will be described in the next section.

Table 5. ScandiNova K2 Modulator Upgrade Specification

ScandiNova K2 Modulator Upgrade Specification		
Parameters	Range	Unit
Peak Power	68.8	MW
Pulse Voltage	250	kV
(Peak Inverse Voltage)	100	kV
Pulse Current	275	A
Average Power	≤ 1	kW
Repetition Rate	1-3	Hz
Pulse Width	4	μ s
Voltage Rise Time	1	μ s
Voltage Fall Time	1	μ s
Voltage Top Flatness (in 3 μ s)	± 0.25	%
Voltage Repeatability (pulse to pulse)	± 0.05	%
Pulse Jitter	$\leq \pm 5$	ns
Modulator Efficiency	>80	%
Klystron Filament Voltage	20	V
Klystron Filament Current - *Note 3	20	A
Klystron warm-up time	60	Minutes

Note 3: When the heater power is applied to a cold tube, the heater voltage shall be adjusted from zero to prescribed value so that the heater current should be kept at 20 A (Not to exceed 20 A! Otherwise, the klystron may be damaged). This value of heater voltage shall be maintained for at least 60 minutes prior to the application of beam voltage. The liquid coolant flow must be operating whenever the heater power is applied.

6.0 OTHER MODULATOR K2 UPGRADE REQUIREMENTS

The K2 upgrade was discussed with ScandiNova experts. They believe the existing modulator can supply 275 A current (10% increase) and 250 kV high voltage to Toshiba klystrons without changing the high voltage DC power supply, the switch units or the main high voltage transformer.

Except for the modulator specification that is upgraded from ITT 8568 to Toshiba klystron E37300, the other requirements will be described in following sections.

6.1 REMOVE CAPACITIVE PLATE IN TRANSFORMER TANK

The capacitive plates that were installed in the oil tank are used to smooth overshoot and improve the pulse top flatness in the old ScandiNova K2 modulators.

In the new model of K2, ScandiNova should improve the waveform by using solid flat copper cables between the switch units and the transformer to reduce the inductance of the cables that will allow removing the existing capacitive plates to get the same waveform quality.

ScandiNova shall remove all capacitive plates from the modulator oil tanks. This will also improve the isolation between the new Toshiba klystron high voltage and the tank wall.

6.2 REPLACE KLYSTRON SOCKET TO ACCOMMODATE E37300

The old ITT klystron sockets cannot accommodate the new Toshiba klystrons. To support the new Toshiba klystrons, the old sockets have to be replaced.

ScandiNova shall work with Toshiba to design and manufacture a new socket for the first klystron that will be installed in July, 2015 in the test modulator. ScandiNova shall be responsible for ensuring successful acceptance of the new Klystron socket design prior to manufacturing the remaining six sockets.

6.3 REPLACE THE CABLES BETWEEN SWITCH UNITS AND TRANSFORMER TANK

The cables supply high voltage from the switch units to the high voltage transformers in the oil tanks. Existing cables are the multi-strand wires with high inductance. The inductances of the wires produce extra overshoot on the waveform. ScandiNova developed a new technology to use solid flat copper cables instead of the multi-strand wires. In that way, the inductance is significantly reduced to improve the waveform. Consequently, the capacitive plates in the oil tank can be removed completely.

The Toshiba klystron length is much longer than the ITT klystron's. In fact, the capacitive plate under the klystron socket will need to be removed in order to allow more space to accommodate the new klystron in the oil tank.

ScandiNova shall design and manufacture the new solid copper cables to replace the old ones.

6.4 REPLACE PULSE TUNING CIRCUITS IN THE OIL TANK

There are thirty (30) switch units supplying power to the high voltage transformer. Each switch channel needs a pulse tuning circuit to improve the pulse waveform flatness that is the most important parameter to the modulator.

All existing tuning circuits have to be replaced by new circuits when the klystrons are replaced.

ScandiNova shall design and manufacture the pulse tuning circuits to fit each modulator and klystron. After the installation, the tuning circuits shall satisfy the requirement of the pulse top flatness $\pm 0.25\%$ in 3 μs .

6.5 REPLACE KLYSTRON TANK LID

To accommodate new Toshiba klystrons and solenoids on the modulator oil tank, the tank lids that support both klystrons and its focus solenoids have to be replaced.

ScandiNova shall work with Toshiba to design new lids to fit modulator tank and to support the solenoid and klystron properly.

6.6 REPLACE THE KLYSTRON FOCUS SOLENOIDS

CLS linac modulators use two (2) electromagnets and five permanent magnets to accomplish electron beam focus in the ITT klystrons.

However the Toshiba klystron uses five (5) coils and one (1) reversed coil to form a solenoid to focus electron beam. So, existing permanent magnets and electromagnets have to be replaced by the specially designed solenoids or electromagnets to supply proper magnetic field for the klystrons.

The magnets shall come with the klystrons from Toshiba. ScandiNova shall install them with the klystrons when assembling the modulators.

6.7 INSTALL THE SOLENOID POWER SUPPLIES

The electromagnets used with the klystrons need power supplies to drive them.

Each electromagnet has six coils that need a six channel power supply to provide proper power.

ScandiNova shall purchase the power supplies from Toshiba and install them at CLS. Each power supply must have two interlock (relay) contacts/terminals with one of them normally open and another one of them normally closed. The power supplies should have readable meters to indicate the voltage and current separately. ScandiNova also shall complete integration of control and monitoring of klystron solenoid power supplies into the modulators.

The power supplies shall be installed in a rack. ScandiNova shall inform CLS of the required sizes of the rack. CLS shall be responsible to find space for the rack. If possible, the power supplies may be mounted in an existing rack as per Modulator Room 1032 – Waveguide Assemblies ACCL\ME\MOD\0184300 [1]. If impossible, ScandiNova shall provide the rack.

6.8 REPLACE THE CTV BOARDS IF NECESSARY

The CTV boards in the transformer oil tank may need to be replaced because the new electromagnets that will be installed are different from both the old electromagnets and the permanent magnets. The CTV boards are used to measure the high voltage on the klystron. ScandiNova shall be responsible for any modification to the location of the CTV boards and any re-calibration that would be required when the new klystrons are installed.

6.9 CONTROL SYSTEM UPGRADE

As discussed with ScandiNova, ScandiNova shall be responsible to make all necessary control change, especially the change for the klystron electromagnet (solenoids) power supplies. Any upgrades shall conform to the CLSI Control System Common Specification [2]. CLS shall be responsible for the control modification to the electromagnet cooling system and new ion pump controller's interlocks.

The software used for modulator #1 and #2 with electromagnets may need to be copied to the test modulator and modulators #3-6. ScandiNova shall reset all applicable interlock levels, e.g. the klystron filament increases from 14 A to 20 A and the cooling water flow changes for the klystron solenoids.

6.10 COVERS

The new covers on the top of the modulators that are used to cover the klystrons have to be redesigned and manufactured. ScandiNova shall re-design and manufacture these covers, then install them in CLS modulators.

This work may be left to the end of the commissioning because the klystron has to be located at the proper position on the modulator. And the old cover also may fit the new tube. CLS and ScandiNova should first consider the possibility of modifying existing cover.

6.11 WATER SYSTEM MODIFICATION

The klystron E37300 current will increase 10% compared with the ITT 8568 klystron. The increment of the power in the modulator may need extra cooling water. ScandiNova shall be responsible for the internal modification of the modulator if required. ScandiNova shall inform CLS how much cooling water is needed to support the new modulator. CLS can complete the mechanical modification to the exterior of the modulator. ScandiNova shall be responsible for providing the detailed requirements to CLS.

CLS shall be responsible for the design, manufacture, installation and testing of the water cooling system for the new klystron and the electromagnet.

6.12 ION PUMP HIGH VOLTAGE POWER SUPPLY

The ion pump in the klystron E37300 shall use Toshiba Ion Pump Power Supply VT-69009. The input voltage available for the power supply is 60Hz 120V. The output voltage of the power supply shall not exceed $3.9\text{kV} \pm 5\%$. The power supply shall interlock the modulator if the output current exceeds a threshold value. CLS shall be responsible for interfacing to the analog current monitor of the power supply for monitoring the vacuum in the klystron.

6.13 SCANDINOVIA SHALL BE RESPONSIBLE FOR ALL NECESSARY UPGRADES

Many devices and parts are not mentioned here that may impact the modulator working with the new Toshiba klystron. E.g. the klystron filament power supply needs power increase from 15V/14A to 20V/20A. It is not mentioned because ScandiNova already confirmed the old klystron filament power supply is enough to provide suitable voltage and current to the new klystron.

ScandiNova shall be responsible to consider all necessary modification, through consultation with Toshiba to ensure the upgrade is able to support the klystron E37300 used in CLS.

Power supplies and components shall have either one of the certification marks listed in http://www.saskpower.com/services/cecr_sk/bulletins/item14.shtml or shall be approved by a Standards Council of Canada accredited testing laboratory to SPE-1000 standard and have a corresponding Special Inspection Label listed in the above bulletin indicating this. A copy of the inspection report shall be provided with the equipment.

Power supply racks, cabinets, enclosures and combinations thereof shall conform to the BKL report "Vibration Isolation Mechanical Equipment" [3].

The first modulator modification shall be completed in July or August, 2015 at the test modulator room in CLS. After commissioning the new modulator with the klystron E37300, the same modification shall be used for the remaining six modulators over the next 2-3 years to complete all modulator upgrades.

7.0 SHIPMENT

All the parts that will be installed (replaced) in CLS shall be delivered to CLS within 5 months of contract signing.

ScandiNova shall be responsible for the shipment in accordance with contract terms.

8.0 APPLICABLE CODES, STANDARDS AND PROCEDURES

The following documents shall be considered as part of this specification. All equipment shall be built in strict accordance with the following standards:

CSA C22.1-12 Canadian Electrical Code, Part I Safety Standards for Electrical Installations

Saskatchewan Interpretations of the Canadian Electrical Code, SaskPower, Version 1.0, 2012

All other applicable CSA standards

IEEE Std 519-2014 Recommended Practices and Requirements for Harmonic Control in Electrical Power Systems

Industry Canada ICES-001 Industrial, Scientific and Medical Radio Frequency Generators

IEEE Std C57 Series for Distribution, Power and Regulating Transformers including but not limited to:

C57.12.01 - 2005

C57.12.56 - 1986

C57.12.59 – 2001

C57.12.80 – 2010

C57.12.91 – 2001

C57.110 – 1998

CSA C22.2 No.47-13 Air-Cooled Transformers (Dry Type)

CSA C9-02 (2007) Dry-Type Transformers

ANSI C37.50 Low-Voltage AC Power Circuit Breakers Used in Enclosures—Test Procedures

CSA/SPE-1000-13 Model code for the Field Evaluation of Electrical Equipment

CAN/CSA – ISO 9001:08

Saskatchewan Occupational Health and Safety Regulations, 1996

Unless otherwise specified, the issue date or revision level shall be that in effect on the date of the Invitation to Quote. Exceptions to these standards shall be reviewed and accepted by CLSI.

9.0 QUALITY ASSURANCE

The ScandiNova shall maintain and apply a quality assurance program compliant with ISO-9001 for the design, manufacture and testing of all components. ScandiNova shall be responsible for solving all detail technical problems to operate the modulator K2 with Toshiba klystron E37300.

10.0 DESIGN REVIEW, INSPECTION TESTING AND COMMISSIONING

CLS staff shall join in the modification of the modulators and the installation of the klystrons.

ScandiNova shall formulate acceptance test procedures for all components and will provide the facility and instrumentation to perform all relevant tests to ensure compliance with this specification. The acceptance test procedures shall include but not be limited to all of the testing procedures specifically outlined in this document. These test procedures are subject to CLS review and acceptance.

CLS reserves the right to have access to the ScandiNova facility at any time during the fabrication and testing procedures. CLS shall have the right to witness any manufacturing or testing procedures upon request. When requested, the ScandiNova shall provide a minimum of 21 days notice in advance of any test date to allow for CLS to make the necessary travel arrangements.

CLS reserves the right to require additional or more extensive tests to be conducted in the event of marginal design or performance.

All items purchased or manufactured by a subcontractor used in the work shall be clearly identified to CLS.

Review and acceptance of CLS shall not release the ScandiNova from its responsibility to correct errors, oversights and omissions to ensure conformance to the specifications in this document.

Isolation testing shall be performed on all power supplies as specified in the Dielectric Voltage Withstand section of SPE-1000.

The modulators shall undergo initial functional testing. This testing shall include but not be limited to the following:

1. Control functions shall be exercised through all states.
2. All interlock functions shall be checked for proper operation and indications.

3. Heat checking shall be done on all critical components to insure operation within manufacturer's ratings. ScandiNova shall temporarily install thermocouples on critical components and on all critical points. These points shall be agreed on between the ScandiNova and CLS. Alternatively a calibrated infra-red imaging camera may be used for these measurements. Measurements shall be taken after the modulator has reached thermal equilibrium.

ScandiNova shall submit certified test reports to CLS for approval before shipping the modulators.

For purpose of warranty under an order for these modulators, final acceptance of installation is defined as the successful completion of acceptance tests at CLS to substantiate the compliance with this specification. Final Acceptance tests shall be completed within three months of the delivery date. ScandiNova shall be notified of the test dates and may have a representative present to witness the tests. ScandiNova shall be held responsible for the contracted performance of the modules produced and delivered to CLS.

Final Acceptance testing shall include any or all of the following tests:

1. Tests described in this section.
2. Connection to actual loads and verification of conformance to specifications.

11.0 RELIABILITY AND MAINTAINABILITY

Each modulator modification shall have a warranty of at least one year after final acceptance.

The modulators shall be designed and constructed for an expected operational lifetime of greater than 20 years. It is understood that maintenance may be required during this period. Subassemblies should be designed for repair rather than replacement.

The modulators shall be designed and constructed to be in continuous use with only limited maintenance periods throughout the year. Major maintenance periods shall be scheduled twice a year and have a duration of approximately 15 working days. Weekly maintenance periods will likely be scheduled to last for one 8 hour time period. Weekly maintenance should not be required. The modulators shall have a minimum expected mean time between failure of 40,000 hours in the operational environment at CLS as outlined in this specification.

The modulators shall be designed and constructed in such a way as to supply access to any system components that are expected to fail in the normal operating environment. Large assemblies that need to be removed from cabinets or racks for service should be mounted on slides. Safe and easy access to all connection points to external devices shall be incorporated into the overall design. Easy access shall be made available to all components, especially solid state power components, fuses, and printed circuit boards. All components shall be removable without cutting or removing cabinet frame members. ScandiNova drawings shall show the positions of the major components in the modulator.

In general, the modulators can be set up and operated according to standard industrial practices. However, maintenance and testing in some cases may be expedited by deviating from standard practices, provided extra precautions are taken to reduce hazards to an acceptable level before these deviations are permitted. If applicable, these precautions must be clearly documented in the manuals.

ScandiNova shall supply a list of recommended spare parts along with cost information.

Different colored wires shall be used wherever possible to facilitate troubleshooting.

Each cable and wire shall be identified using mechanically printed heat shrink markers at each end. These shall be keyed to identical numbers used on electrical schematics.

All threaded mechanical fasteners should be American Standard Thread. The use of metric standards is negotiable with CLS. If metric standards are used, spare nuts, bolts and washers shall be provided by ScandiNova.

12.0 OTHER REQUIRMENTS AND CONSTRAINTS

All labelling, manuals, and other documentation shall be in English.

ScandiNova shall supply complete documentation of all components and subsystems. Hardcopies (2) and electronic copies (including editable source files) of all documentation shall be provided.

All documentation should conform to the "Vendor Documentation Specification". [4] This documentation shall include but be not limited to the following:

Users/operation/maintenance manuals, which shall contain, but not be limited to:

1. specifications, operating instructions, control system interface commands/protocols, hook-up instructions, circuit diagrams, block diagrams, maintenance procedures (both operational and preventative), component data sheets, normal adjustments and calibration setup procedures.
2. A final parts list along with a list of recommended spare parts and their cost information.
3. A complete set of electrical and mechanical drawings shall be supplied for each modulator.

ScandiNova shall provide CLS with electronic copies of all custom developed source code, PLC programs and embedded logic in source and binary format on CD-ROM.

All test measurement results shall be provided in hardcopy, on CD-ROM, and archived at ScandiNova's location for at least the duration of the warranty.

Each modulator shall have a stainless steel or aluminum nameplate on each unit that shall include the following information:

CLS Equipment Tag number (Details to be worked out with CLS)
Manufacturer's name and address
Model name/number
Serial number
Input voltage and current ratings
Output voltage and current ratings
Gross weight of the unit
Date of manufacture
Cooling requirements (if required)

The nameplate information should be in 14 point size fonts.

All shipping containers shall be marked or tagged with the following information:

CLS purchase order number.
Shipping address as specified within the contract.
ScandiNova's name.
Components contained within each package.
"Top-side up" if required.
"Fragile" if required.
Shock and Tip sensors on containers are encouraged.

Where possible shipping should be by air cushion truck. Shipping by rail or flatbed truck is discouraged.

ScandiNova shall notify CLS ten (10) days prior to shipment. The notice shall include a list of equipment, origin, shipping date, shipping firm, routing of shipment, anticipated date of arrival, and shipping weight. CLS reserves the right to approve shipping firm and route.

13.0 DRAWINGS AND DOCUMENTATION

All relevant as-built mechanical and electrical drawings from the modifications necessary to support the equipment installation, operation and maintenance shall be submitted to CLS after the commissioning.

All documentation shall conform to the vendor documentation requirements for the Canadian Light Source Inc. and be submitted to CLS.

14.0 REFERENCES

- [1] Modulator Room 1032 – Waveguide Assemblies ACCL\ME\MOD\0184300
- [2] "Control System Common Specification" 7.4.39.1
- [3] "Vibration Isolation Mechanical Equipment", BKL report
- [4] "Vendor Documentation Specification" 0.4.1.1

REVISION HISTORY

<i>Revision</i>	<i>Date</i>	<i>Description</i>	<i>Author</i>
A	2014-Dec-23	Original Draft	Xiaofeng Shen
0	2015-Feb-02	Incorporated comments from review. Issued for use.	Xiaofeng Shen
1	2015-Feb-05	Modulator Room Drawing # has been fixed.	Xiaofeng Shen